

### Assignment #6 - Date: 17/2/26

Consider a hospital having several departments (like cardiology, medicine, neurology, etc.). Each doctor can belong to only one department but a department can have multiple doctors. A patient may be admitted under more than one department and be assigned to one or more doctors of different departments. There is a date of admission and a date of release for each patient. A patient is identified by a Patient\_Id. For each patient, we also maintain information like first name, last Name, address (comprised of house no., area, pincode and state), gender, date of birth, and one or more phone numbers. Every doctor has Doctor\_Id, name, date of birth, department and highest degree. A department is identified by its department\_code. Other attributes of a department are department name, phone number, name of the HoD and number of wards.

1. First draw the ER diagram and derive the tables using correct rules for relational model generation from the ER diagram.
2. Execute the following **queries** based on the problem stated above.
  - a. List the patients (Patient\_Id, Name) so far admitted under each doctor (Doctor\_Id, Name) who have not been admitted in the same department to which the doctor belongs.
  - b. List the patients (Patient\_Id, Name) so far admitted under each doctor (Doctor\_Id, Name) who have been admitted under more than one department.
  - c. List the patients (Patient\_Id, Name) currently admitted under each doctor (Doctor\_Id, Name), i.e., the patients who have been admitted but released after one month.
  - d. List the department(s) (Names) having maximum number of patients.
  - e. List the doctors (Doctor\_Id, Name) who never treated any patient from Neurology.
  - f. Patients (Patient\_Id, Name) who were never admitted under the HoD's department.
  - g. Find patients (Patient\_Id, Name) admitted in only one department (exists/not exists)
  - h. Find patients (Patient\_Id, Name) treated by every doctor in a department (exists/not exists)
  - i. Find youngest patient (Patient\_Id, Name) using ALL
  - j. Search for doctors (Doctor\_Id, Name) whose department has more patients than any other department (using any)